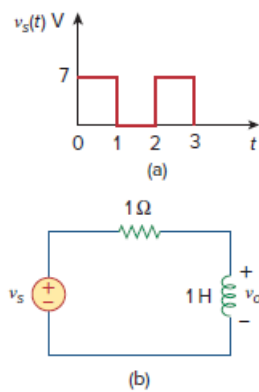


Problem

If the square wave shown in Fig. 17.74(a) is applied to the circuit in Fig. 17.74(b), find the Fourier series for $v_o(t)$.

Figure 17.74:



Step-by-step solution

Step 1 of 8

Refer to Figure 17.74 in the textbook.

The source voltage $v_s(t)$ is defined as,

$$v_s(t) = \begin{cases} 1 & 0 < t < 1 \\ 0 & 1 < t < 2 \end{cases}$$

Consider the time period $T = 2$.

Determine the fundamental frequency ω_o .

$$\begin{aligned} \omega_o &= \frac{2\pi}{T} \\ &= \frac{2\pi}{2} \\ &= \pi \text{ rad/sec} \end{aligned}$$

Step 2 of 8

Determine the Fourier series of source voltage.

Consider the general equation as,

$$v_s(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)) \dots\dots (1)$$

Determine the value of a_0 .

$$\begin{aligned} a_0 &= \frac{1}{T} \int_0^T v_s(t) dt \\ &= \frac{1}{T} \left[\int_0^1 v_s(t) dt + \int_1^2 v_s(t) dt \right] \\ &= \frac{1}{2} \left[\int_0^1 1 dt + \int_0^1 0 dt \right] \\ &= \frac{1}{2} \left[\int_0^1 1 dt \right] \end{aligned}$$

$$\begin{aligned} a_0 &= \frac{1}{2} [t]_0^1 \\ &= \frac{1}{2} [1 - 0] \\ &= \frac{1}{2} [1] \end{aligned}$$

$$a_o = \frac{1}{2} \dots\dots (2)$$

Step 3 of 8

Consider,

$$\begin{aligned} a_n &= \frac{2}{T} \int_0^T v_s(t) \cos n\omega_0 t dt \\ &= \frac{2}{T} \left[\int_0^1 v_s(t) \cos n\omega_0 t dt + \int_1^2 v_s(t) \cos n\omega_0 t dt \right] \\ &= \frac{2}{2} \left[\int_0^1 (1) \cos n\pi t dt + \int_1^2 (0) \cos n\pi t dt \right] \\ &= (1) \left[1 \int_0^1 \cos n\pi t dt \right] \end{aligned}$$

$$\begin{aligned} a_n &= (1) \frac{\sin n\pi t}{n\pi} \Big|_0^1 \\ &= \left(\frac{1}{n\pi} \right) [\sin n\pi(1) - \sin n\pi(0)] \\ &= \left(\frac{1}{n\pi} \right) [\sin n\pi - 0] \end{aligned}$$

$$\begin{aligned} a_n &= \left(\frac{1}{n\pi} \right) [\sin n\pi] \\ &= \left(\frac{1}{n\pi} \right) [0] \end{aligned}$$

$$a_n = 0 \dots\dots (3)$$

Step 4 of 8

Determine the value of b_n .

$$\begin{aligned}
 b_n &= \frac{2}{T} \int_0^T v_s(t) \sin n\pi t \, dt \\
 &= \frac{2}{T} \left[\int_0^1 v_s(t) \sin n\pi t \, dt + \int_1^2 v_s(t) \sin n\pi t \, dt \right] \\
 &= \frac{2}{2} \left[\int_0^1 (1) \sin n\pi t \, dt + \int_1^2 (0) \sin n\pi t \, dt \right] \\
 &= (1) \left[\int_0^1 \sin n\pi t \, dt \right] \\
 b_n &= (1) \left. \frac{-\cos n\pi t}{n\pi} \right|_0^1 \\
 &= -\left(\frac{1}{n\pi} \right) [\cos n\pi(1) - \cos n\pi(0)] \\
 &= -\left(\frac{1}{n\pi} \right) [\cos n\pi - 1] \\
 &= \left(\frac{1}{n\pi} \right) [1 - \cos n\pi]
 \end{aligned}$$

Step 5 of 8

Consider $\cos(n\omega) = (-1)^n$.

$$\begin{aligned}
 b_n &= \left(\frac{1}{n\pi} \right) [1 - (-1)^n] \\
 b_n &= \begin{cases} \frac{2}{n\pi}, & n = \text{odd} \\ 0, & n = \text{even} \end{cases} \dots (4)
 \end{aligned}$$

Substitute equation 2,3,4 in equation 1.

$$\begin{aligned}
 v_s(t) &= \frac{1}{2} + \frac{2}{\pi} \sin \pi t + \frac{2}{3\pi} \sin 3\pi t + \frac{2}{5\pi} \sin 5\pi t + \dots \\
 v_s(t) &= \frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{1}{n} \sin n\pi t, \quad n = 2k-1 \dots (5)
 \end{aligned}$$

Step 6 of 8

Determine the Fourier series of $v_o(t)$.

Apply voltage division.

$$\begin{aligned}
 \mathbf{V}_o &= \left[\frac{j\omega_n}{j\omega_n + 1} \right] \mathbf{V}_s \\
 \mathbf{V}_o &= \left[\frac{jn\pi}{1 + jn\pi} \right] \mathbf{V}_s \\
 \mathbf{V}_s &= \frac{2}{n\pi} \angle -90^\circ \text{ V} \dots (6)
 \end{aligned}$$

For dc component $n = 0$.

Replace 0 for n in equation 5.

$$\mathbf{V}_s = 0.5 \text{ V}$$

Step 7 of 8

Replace 0.5 V for $\mathbf{V}_s$ in equation 6.

$$\mathbf{V}_{dc} = \left[\frac{j(0)\pi}{1+j(0)\pi} \right] \frac{1}{2}$$

$$= 0$$

The inductor is a short circuit to dc. For n^{th} harmonic,

$$\mathbf{V}_s = \frac{2}{n\pi} \angle -90^\circ \text{ V}$$

$$\mathbf{V}_{nth} = \left[\frac{jn\pi}{1+jn\pi} \right] \left(\frac{2}{n\pi} \angle -90^\circ \right)$$

$$= \left[\frac{jn\pi}{1+jn\pi} \right] \left(\frac{2}{n\pi} \angle -90^\circ \right)$$

$$= \left[\frac{n\pi \angle 90^\circ}{\sqrt{1^2 + (n\pi)^2} \angle \tan^{-1} \left(\frac{n\pi}{1} \right)} \right] \left(\frac{2}{n\pi} \angle -90^\circ \right)$$

$$= \frac{2}{\sqrt{1+n^2\pi^2} \angle \tan^{-1}(n\pi)}$$

$$\mathbf{V}_{nth} = \frac{2 \angle -\tan^{-1}(n\pi)}{\sqrt{1+n^2\pi^2}}$$

Step 8 of 8

Consider the equation for $\mathbf{V}_o$.

$$\mathbf{V}_o = \mathbf{V}_{dc} + \mathbf{V}_{nth}$$

$$= 0 + \frac{2 \angle -\tan^{-1}(n\pi)}{\sqrt{1+n^2\pi^2}}$$

$$\mathbf{V}_o = \frac{2 \angle -\tan^{-1}(n\pi)}{\sqrt{1+n^2\pi^2}} \dots\dots (7)$$

Write the equation 7 in time domain.

$$v_o(t) = \sum_{k=1}^{\infty} \frac{2}{\sqrt{1+n^2\pi^2}} \cos(n\pi t - \tan^{-1}(n\pi)) \quad n = 2k-1$$

Therefore, the voltage across inductor is

$$v_o(t) = \sum_{k=1}^{\infty} \frac{2}{\sqrt{1+n^2\pi^2}} \cos(n\pi t - \tan^{-1}(n\pi)) \quad n = 2k-1.$$